ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

# INTRODUCTION TO SYSTEMS DEVELOPMENT

## PAPER 1 · PRACTICE A · MARKING GUIDELINE

**MARKS: 100**

**Marking guideline for the Academic Central NC(V) Level 2 Introduction to Systems Development paper 1 · practice a.**

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 2. It is not a Department of Higher Education and Training marking guideline.

**How to mark:** each ✓ is one mark. Accept answers in the student's own words if the meaning is correct. Accept any correct pseudocode style or flowchart that gives the right result; do not penalise small differences in keywords.

## SECTION A

## QUESTION 1

**1.1.1** D ✓
**1.1.2** A ✓
**1.1.3** C ✓
**1.1.4** C ✓
**1.1.5** A ✓
**1.1.6** D ✓
**1.1.7** B ✓
**1.1.8** D ✓
**1.1.9** B ✓
**1.1.10** C ✓

(10)

**1.2.1** False ✓
**1.2.2** True ✓
**1.2.3** True ✓
**1.2.4** False ✓
**1.2.5** True ✓
**1.2.6** True ✓
**1.2.7** False ✓
**1.2.8** False ✓
**1.2.9** True ✓
**1.2.10** True ✓

(10)

**1.3.1** F ✓
**1.3.2** D ✓
**1.3.3** J ✓
**1.3.4** G ✓
**1.3.5** E ✓
**1.3.6** H ✓
**1.3.7** B ✓
**1.3.8** C ✓
**1.3.9** A ✓
**1.3.10** I ✓

(10)

**TOTAL SECTION A: [30]**

## SECTION B

## QUESTION 2

**2.1** Planning (preliminary investigation) ✓; analysis ✓; design ✓; implementation (development, testing and conversion) ✓; maintenance (review / support) ✓ – in this order (5)

**2.2** Study the current paper system ✓; gather facts from the owners, drivers and marshals (interviews, questionnaires, observation) ✓; identify problems and the requirements of the new system ✓. (3)

**2.3** Any TWO with an advantage: interview ✓ – can ask follow-up questions / get detailed answers ✓; questionnaire ✓ – reaches many people quickly and cheaply / anonymous ✓; observation – see how work is really done; document review – shows actual data used (4)

**2.4** Direct changeover: the old system stops and the new system starts on the same day ✓ – cheap and fast but risky ✓. Parallel running: old and new systems run together for a period ✓ – safe, results can be compared, but double work and costly ✓. Recommendation with reason, e.g. parallel running, because daily takings (money) must not be lost if the new system fails ✓ (accept direct with a sensible reason) (5)

**2.5** Any THREE: to fix errors (bugs) found after implementation ✓; to adapt to new requirements, laws or tax rates ✓; to improve performance or add features ✓; to keep up with new hardware or operating systems; security updates (3)

**[20]**

## QUESTION 3

**3.1** Example (✓ for each part):

```
INPUT amount                          ✓
IF amount > 50 THEN                   ✓
   discount ← amount * 0.1            ✓
ELSE
   discount ← 0                       ✓
ENDIF
toPay ← amount - discount             ✓
OUTPUT "Amount to pay: R", toPay      ✓
```

(6)

**3.2** Oval Start ✓ → parallelogram Input amount → diamond "amount > 50?" ✓ with Yes/No branches labelled ✓ → rectangle discount = amount × 0,1 (Yes) and discount = 0 (No) ✓ → rectangle toPay = amount − discount ✓ → parallelogram Output toPay → oval Stop ✓ (correct symbols and flow lines) (6)

**3.3** Input: amount ✓; Processing: test if amount > 50, calculate discount and amount to pay ✓; Output: amount to pay ✓ (3)

**[15]**

## QUESTION 4

**4.1** Sensible name (camelCase or similar, no spaces) and type, 1 + 1 each: (a) numStudents ✓ – integer ✓ (b) breadPrice ✓ – real / float / currency ✓ (c) hasPaid ✓ – Boolean ✓ (6)

**4.2** (a) 7 + 6 = 13 ✓ (b) 10 / 2 = 5 ✓ (c) 1 ✓ (d) 2 ✓ (4)

**4.3** 4.3.1 Trace table (✓ per correct row, ✓✓ for correct final values):

| i | mark | count | total |
|---|---|---|---|
| – | – | 0 | 0 |
| 1 | 60 | 1 | 60 |
| 2 | 35 | 1 | 95 |
| 3 | 80 | 2 | 175 |
| 4 | 45 | 2 | 220 |

(6)

4.3.2 55 ✓ and 2 ✓ (average 220 / 4 = 55; count 2) (2)

4.3.3 The number of marks of 50 or more (number of students who passed) ✓ (1)

4.3.4 Example:

```
i ← 1                                  ✓
WHILE i <= 4                           ✓
   INPUT mark
   IF mark >= 50 THEN count ← count + 1
   total ← total + mark                ✓ (body unchanged)
   i ← i + 1                           ✓
ENDWHILE
```

(4)

4.3.5 Selection: lines 5–7 ✓; iteration: lines 3–9 ✓ (2)

**[25]**

## QUESTION 5

**5.1** Normal: any age from 16 to 34, e.g. 20 ✓; boundary: 15 or 35 ✓; erroneous: e.g. 40, –3, 12 or "abc" ✓ (3)

**5.2** Syntax error: breaks the rules (grammar) of the language, so the program will not run / compile ✓, e.g. a misspelt keyword such as PRNT or a missing bracket ✓. Logic error: the program runs but gives wrong results ✓, e.g. using + instead of − or > instead of >= ✓. (4)

**5.3** Any TWO: user documentation (user manual / help files) ✓ – used by end users to learn how to use the system ✓; technical documentation (program code comments, data dictionary, system design) – used by programmers / maintenance staff ✓ (1 for each type, 1 for users; max 3) (3)

**[10]**

**TOTAL SECTION B: [70]**

**TOTAL: 100**

**END OF MARKING GUIDELINE**